Unit 1. Introduction

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1.1Artificial Intelligence

What is intelligence?

Intelligence is:

– the ability to reason

– the ability to understand

– the ability to create

– the ability to Learn from experience

– the ability to plan and execute complex tasks.

What is Artificial Intelligence?

• Barr and Feigenbaum “Artificial Intelligence is the part of computer science concerned with designing intelligence computer systems, that is, systems that exhibit the characteristics we associate with intelligence in human behavior.”

• Elaine Rich “AI is the study of how to make computers do things at which, at the moment, people are better.”
• Russell and Norvi “The study of agent that receives percepts from the environment and performs actions.”
• John McCarthy “The science and engineering of making intelligent machines, especially intelligent computer programs.”

So, Artificial Intelligence (AI) is the part of computer science that deals with study and creation of computer systems that can perceive, reason and act. The primary aim of AI is to produce intelligent machine. AI is an interdisciplinary field that requires knowledge in computer science, linguistics, psychology, biology, philosophy and so on.
OR
Artificial intelligence (AI) is a branch of computer science that enables machines to simulate human intelligence, learning, and decision-making to perform tasks that typically require human cognition. These systems use data and algorithms to reason, solve problems, understand language, and make predictions

Different definitions of AI are given by different books/writers. These definitions can be divided into two dimensions.

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

- Top dimension is concerned with thought process and reasoning, whereas bottom dimension addresses the behavior.
- The definition on the left measures the success in terms of fidelity of human performance, whereas definitions on the right measure an ideal concept of intelligence which is called rationality.

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1.2Hard vs. Strong AI, Soft vs. Weak AI:

2 1. Hard AI (also called Strong AI)

- Hard Ai

Hard AI refers to Artificial General Intelligence (AGI) — machines that can truly think, understand, and reason like humans across any task.

Hard AI, often synonymous with Artificial General Intelligence (AGI) or "strong AI," refers to theoretical future systems capable of mimicking human cognitive abilities to learn, reason, and apply intelligence across diverse, unstructured, and novel tasks without specific reprogramming. Unlike today's narrow AI, a hard AI system could seamlessly transfer knowledge between unrelated fields—such as applying principles from physics to solve a complex legal problem—and display adaptability to unexpected scenarios.

Real-world examples in development, often categorized as AGI-hard problems, include autonomous robotic assistants capable of navigating unfamiliar kitchens to prepare food, or advanced AI agents that could, for instance, autonomously identify, chain, and exploit software vulnerabilities at a human level

- Strong AI

Strong AI is the same as Hard AI — it aims to create systems with real consciousness and self-awareness, not just task-specific intelligence.

Strong AI, or **Artificial General Intelligence (AGI)**, is a hypothetical form of AI that possesses human-level intelligence, self-awareness, and the ability to solve a wide range of tasks autonomously. Unlike today's "narrow" AI, which specializes in specific tasks, Strong AI can reason, learn, and adapt across diverse domains without being pre-programmed.

- Soft AI (also called Weak AI)

Soft AI focuses on building systems that can simulate human behavior for specific tasks, without actual understanding or consciousness.

Soft AI, or Narrow AI, refers to computer systems designed to perform specific tasks or solve particular problems within a limited scope, operating under set constraints rather than possessing human-like general intelligence.

Real-world examples include virtual assistants like Apple's Siri or Amazon's Alexa, which interpret voice commands to set timers or play music. Another example is the recommendation algorithms on streaming platforms such as Netflix, which suggest content based on previous viewing habits.

- Weak AI

Weak AI is the same as Soft AI — it is designed for narrow applications, like voice recognition, and cannot generalize beyond its programmed task.

Weak AI, also known as narrow AI, refers to artificial intelligence systems specifically designed and trained to perform a singular, defined task—such as speech recognition or data analysis—without possessing genuine consciousness or broad cognitive abilities. These systems operate under pre-defined constraints and cannot apply their intelligence to tasks outside their specialized domain

. Real-world examples as of 2026 include virtual assistants like Apple's Siri or Amazon's Alexa, recommendation engines on streaming platforms like Netflix, and facial recognition systems used for security.

In summary; **Summary:**

- **Hard AI = Strong AI** → General, human-like intelligence
- **Soft AI = Weak AI** → Narrow, task-specific intelligence

Give short notes on Hard Vs Strong AI

"Hard" AI, also known as Strong AI or Artificial General Intelligence (AGI), is a hypothetical type of AI that would possess the ability to perform any intellectual task that a human being can. In contrast, "soft" AI, or Narrow AI, is designed for specific tasks or a narrow range of tasks. Strong AI aims for a machine with general intelligence and the ability to learn, reason, and adapt like humans, while soft AI focuses on specialized functionality and high accuracy within its defined scope.

Elaboration:

Hard/Strong AI (AGI):

- **Definition:**

  Artificial general intelligence (AGI) is the concept of a machine with general intelligence, mimicking human cognitive abilities.
- **Capabilities:**

  AGI would have the ability to learn, reason, solve problems, and adapt to new situations, just like a human.
- **Example:**

  A hypothetical AI that could understand and respond to a wide range of human questions, learn new skills, and adapt to changing environments.
- **Theoretical:**

  While AGI is a theoretical goal, it is not currently a reality.
- **Challenges:**

  Replicating human-level intelligence and consciousness presents immense challenges for AI researchers.

Give short notes on Soft Vs Weak AI:

"Soft" and "weak" AI are essentially synonymous terms in the field of artificial intelligence (AI), both referring to AI systems that are designed to perform specific tasks or within a narrow range of tasks. This contrasts with the hypothetical "strong" or "general" AI that would possess human-level intelligence and be able to perform any intellectual task.

Here's a more detailed breakdown:

Soft AI/Weak AI:

- **Definition:**
  Also known as narrow AI, this type of AI is designed for specific tasks or a limited set of tasks.
- **Capabilities:**
  It excels at its designated tasks, often achieving high accuracy and efficiency within that defined scope.
- **Examples:**
  Virtual assistants like Siri and Alexa, speech recognition systems, self-driving cars, and recommendation engines are all examples of soft/weak AI.
- **Limitations:**
  It lacks general intelligence and cannot adapt or learn beyond its programmed parameters.

In essence, soft/weak AI simulates intelligence for specific tasks, while strong/general AI would involve true, human-level understanding and adaptability.

NOTE: Is Hard AI only a theoretical concept?

➔ Yes, **hard AI**—more commonly referred to in the industry as **Artificial General Intelligence (AGI)** or **Strong AI**—is currently considered a theoretical concept, as it has not yet been achieved. While artificial intelligence

is a reality today, all current systems are considered **"Narrow AI"** (or Weak AI), designed to excel at specific, limited tasks.

2.1 Foundations and Applications:

Artificial intelligence (AI) is a broad field with deep foundations in computer science, mathematics, and philosophy. It encompasses various techniques like machine learning, deep learning, and artificial neural networks, which enable computers to perform tasks that typically require human intelligence. These techniques are used in diverse applications, from image and speech recognition to autonomous vehicles and virtual assistants.

Foundations of Artificial Intelligence:

➔ **Computer Science:**

- Provides the tools and algorithms needed to implement AI systems, including programming languages, data structures, and computational theory.

➔ **Mathematics:**

- Underlies the algorithms used in AI, with areas like linear algebra, probability theory, and statistics being crucial.

➔ **Philosophy:**

- AI's conceptual foundations are rooted in philosophical inquiries about the mind, reasoning, and knowledge, as explored by thinkers like Plato and Aristotle.

➔ **Neuroscience:**

- Studies of the human brain have influenced AI's development, particularly in the design of artificial neural networks.

➔ **Control Theory and Cybernetics:**

Cybernetics is the study of how systems control themselves and communicate, especially how they respond to changes in their environment to keep working properly. It's like understanding how machines, people, or even animals adjust their actions based on feedback, such as a thermostat turning on the heater when it's too cold.

Control theory is a part of cybernetics that focuses on how to design systems (like robots, cars, or industrial machines) so they automatically stay on track and do what they're supposed to—even when things around them change.

In simple terms, cybernetics and control theory help us build smart systems that can make decisions, learn, and adapt—just like artificial intelligence does

Applications of Artificial Intelligence:

AI applications are software programs leveraging technologies like machine learning, computer vision, and (Natural language Processing) NLP to perform complex tasks, such as content creation, predictive analytics, and process automation. They are widely used for enhancing efficiency in industries like finance (fraud detection), healthcare (diagnostics), and e-commerce (recommendations).

Key AI Applications Across Industries

➔ **Generative AI & Content Creation:**

- Tools like ChatGPT, Claude, and Adobe Firefly create text, code, images, and video, accelerating content production.

➔ **Customer Service & Support:**

- AI-powered chatbots and virtual assistants (e.g., Tidio) provide instant, 24/7 customer support, increasing efficiency.

➔ **Healthcare:**

- AI aids in medical image analysis, drug discovery, and diagnostics, significantly improving speed and accuracy.

➔ **Finance & Security:**

- Machine learning algorithms detect fraudulent transactions in real-time, while AI enhances cybersecurity by identifying threats.

➔ **Marketing & E-commerce:**

- Personalized recommendations (e.g., Netflix, Spotify) and tailored marketing content are driven by AI analyzing consumer behavior.

➔ **Productivity & Workflow:**

- Tools now exist to create applications by simply describing them in natural language, enabling rapid, no-code development.

➔ **Logistics & Transportation:**

- AI optimizes supply chains, manages inventory, and powers autonomous vehicles and navigation system.

➔ **Energy:**

- AI is being used to improve energy efficiency and predict energy demand.

➔ **Government:**

- AI is being used to improve public safety, detect crime, and provide citizen services.

2.2 Intelligent Agent: =>

2.2.1 Introduction of agents: =>

➔ Agents:

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators/effectors.
- A human agent has eyes, ears and others for sensors. Similarly, hands, legs, mouth and other body parts for effectors.
- A robotic agent substitutes cameras and infrared range finders for the sensors and various motors for the effectors.

Agent and Environment

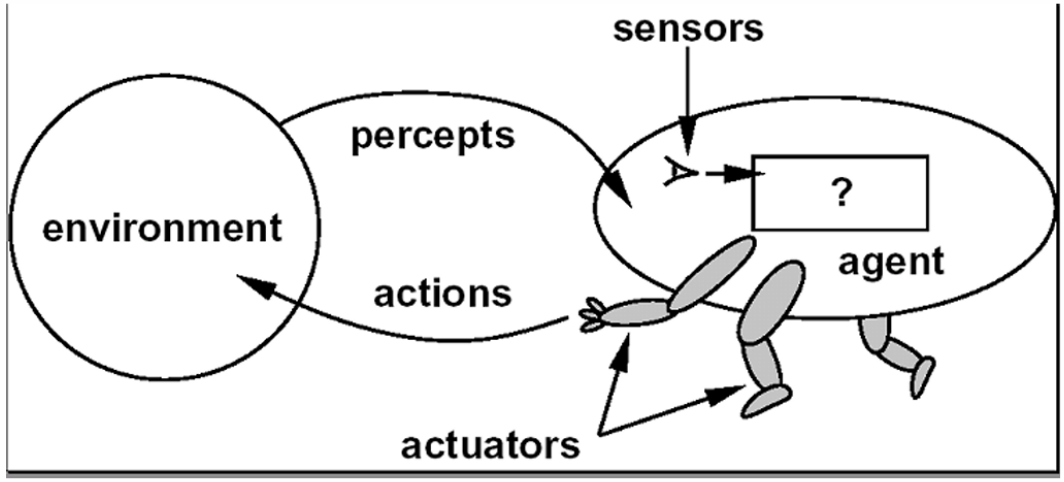


Figure: Agent and Environment

Agents • Operates in an environment • Perceives its environment through sensors. • Acts upon its environment through actuators/effectors • Have goals

Intelligent Agents • Intelligent Agent ✓ Must sense ✓ Must act ✓ Must be autonomous (to some extent) ✓ Must be rational.

1.4.2. Structure of Intelligent agent:

Structure of Agent

➔ Agent Function:

The agent function maps from percept sequences to actions:

F: P* → A

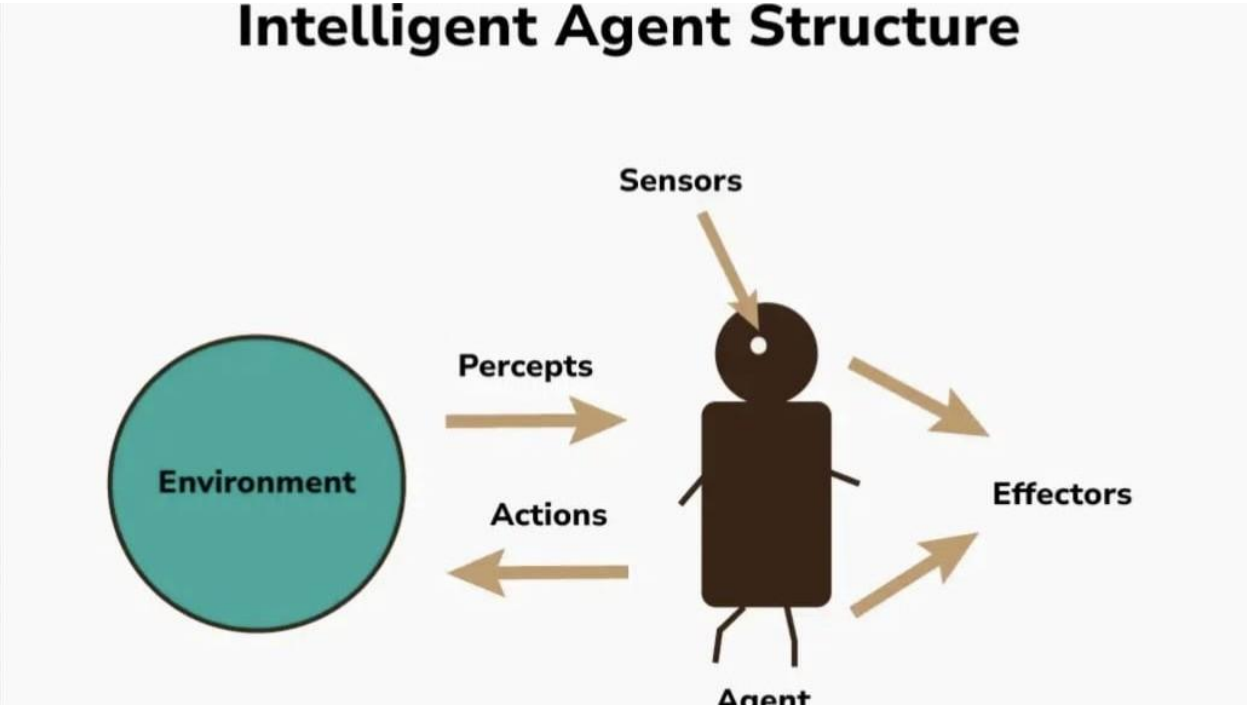
- Where p* is the complete set of percept sequence and A is the action
- The term percept is used to the agents perceptional inputs at any given instant.

An intelligent agent's structure typically involves a continuous cycle of sensing, perceiving, thinking, deciding, and acting. This is achieved through key components like sensors for input, actuators for output, and a decision-making mechanism that considers knowledge, reasoning, and learning. The agent also interacts with its environment, adapting and improving its performance over time.

Here's a more detailed breakdown:

Key Components:

- Environment: The external world the agent interacts with.
- Sensors (Percepts): Input devices that allow the agent to receive information from the environment.
- Actuators: Output devices that allow the agent to interact with the environment.
- Decision-Making Mechanism: This component includes:
  - Knowledge Base: Stored information and experiences.
  - Reasoning: The process of using knowledge to make inferences and decisions.
  - Learning: The ability to adapt and improve performance based on new experiences.
- Action Module: This module translates the agent's decisions into actions within the environment.



Functions of an Intelligent Agent:

- Perceiving: Using sensors to gather information from the environment.
- Processing: Using knowledge and reasoning to interpret perceptions and make decisions.
- Acting: Using actuators to interact with the environment based on decisions.

All modules **work together** to make the agent smart and able to operate.

[[[[NOTE:2. An **actuator** is the physical device that **produces movement or action**, while an **effector** is the **overall system or part** (including the actuator) that **actually causes change** in the environment. Example: In a robot arm:

- The **actuator** is the **motor** that spins and moves parts.
- The **effector** is the **robot hand** that **grabs** things using that movement.]]]]]

Agent Architecture:

An intelligent agent in AI architecture comprises sensors, actuators, and a "brain" (agent program) to perceive, reason, and act autonomously to achieve goals. The architecture structure includes sensors to gather data, memory to recall past states, a planning module for decisions, and actuators to take action in the environment.

Key Components of Agent Architecture

- ➔ **Sensors:**
  - Gather raw data from the environment (cameras, microphones, APIs).
- ➔ **Actuators:**
  - Execute decisions made by the agent (robotic arms, screen displays, API calls).
- ➔ **Agent Program (Brain):**
  - The core logic, ranging from simple rule-based systems to complex LLM-based reasoning.
- ➔ **Memory:**
  - Stores past experiences, observations, and actions to improve future decision-making.
- ➔ **Planning/Reasoning:**
  - Evaluates situations, weighs alternatives, and breaks down complex goals.

Real-World Example: Self-Driving Car

A self-driving car represents a sophisticated hybrid agent architecture combining reflex actions with long-term planning.

➔ **Sensors:**

- Cameras, LIDAR, radar, and GPS, providing real-time data on surroundings, road conditions, and location.

➔ **Architecture (The Brain):**

- **Reflex/Perception System:** Instantly detects obstacles and triggers braking or swerving.
- **Planning System:** Calculates the best route to the destination while adhering to traffic rules.
- **Memory Module:** Stores map data and learns from previous trips to predict traffic patterns.

➔ **Actuators:**

- Steer the wheels, manage acceleration, and operate braking systems.

➔ **Action:**

- The car continuously perceives, plans, and navigates autonomously

1.4.3. Properties of Intelligent Agents

• Intelligent Agent

✓ Must sense

✓ Must act

✓ Must be autonomous (to some extent)

✓ Must be rational

➔ **The properties of intelligent agents are:**

- **Goals**:

  An intelligent agent operates to achieve specific objectives or desired outcomes.

- **Autonomy**:
  An intelligent agent can make decisions and act independently without human intervention.
- **Perception**:
  An intelligent agent senses and gathers information from its environment.
- **Decision-making**:
  An intelligent agent selects appropriate actions based on its perceptions and goals.
- **Learning**:
  An intelligent agent improves its performance or behavior over time through experience.
- **Reactivity**:
  An intelligent agent responds promptly to changes or stimuli in its environment.
- **Action**:
  An intelligent agent performs operations to affect the environment and fulfill its goals.
- **Communication**:
  An intelligent agent can exchange information with other agents or humans.
- **Reasoning**:
  An intelligent agent uses logic and knowledge to infer conclusions and make decisions.
- **Adaptability**:
  An intelligent agent adjusts its strategies or behavior according to new conditions.
- **Adaptive learning**:
  An intelligent agent learns from interactions and modifies its actions to improve future performance.

- **Sociability**:
  An intelligent agent interacts and cooperates with other agents or humans to achieve complex tasks.
- **Utility maximization**:
  An intelligent agent aims to choose actions that maximize a defined performance or utility measure.

1.4.4. PEAS description of Agents :=>

PEAS stands for:
Performance measure, **E**nvironment, **A**ctuators, **S**ensors
It is a framework used to **describe an intelligent agent** completely — covering what it aims for, where it works, how it acts, and how it senses.

PEAS Components:

Component	Meaning	Example
Performance measure	How we evaluate if the agent is successful	High driving safety, speed, fuel efficiency for a self-driving car
Environment	Where the agent operates	Roads, traffic, pedestrians for a self-driving car

Component	Meaning	Example
Actuators	What the agent uses to act on the environment	Steering wheel, accelerator, brakes in a self-driving car
Sensors	What the agent uses to perceive the environment	Cameras, radar, GPS, lidar in a self-driving car

Example for a few agents:

1. **Self-Driving Car**
- Performance measure: Safe driving, obey traffic laws, passenger comfort
- Environment: Roads, traffic, pedestrians, weather
- Actuators: Steering, acceleration, brakes, horn
- Sensors: Cameras, lidar, radar, GPS

2. **Chess-Playing Agent**
- Performance measure: Winning the game, minimizing opponent's winning chances
- Environment: Chessboard, opponent moves
- Actuators: Moving pieces on the board
- Sensors: Vision system (board state), input from opponent

3. **Automated Vacuum Cleaner (Roomba)**
- Performance measure: Cleanliness of the floor, cleaning time, battery usage
- Environment: House rooms, furniture, dust
- Actuators: Wheels, vacuum, brushes

- Sensors: Dirt sensors, bump sensors, cliff sensors

In short:

PEAS helps you define an agent completely by specifying what success looks like, where it acts, how it acts, and how it senses

1.4.5. Types of Agents: Simple Reflexive, Model Based, Goal Based, Utility Based, Learning agent :=>

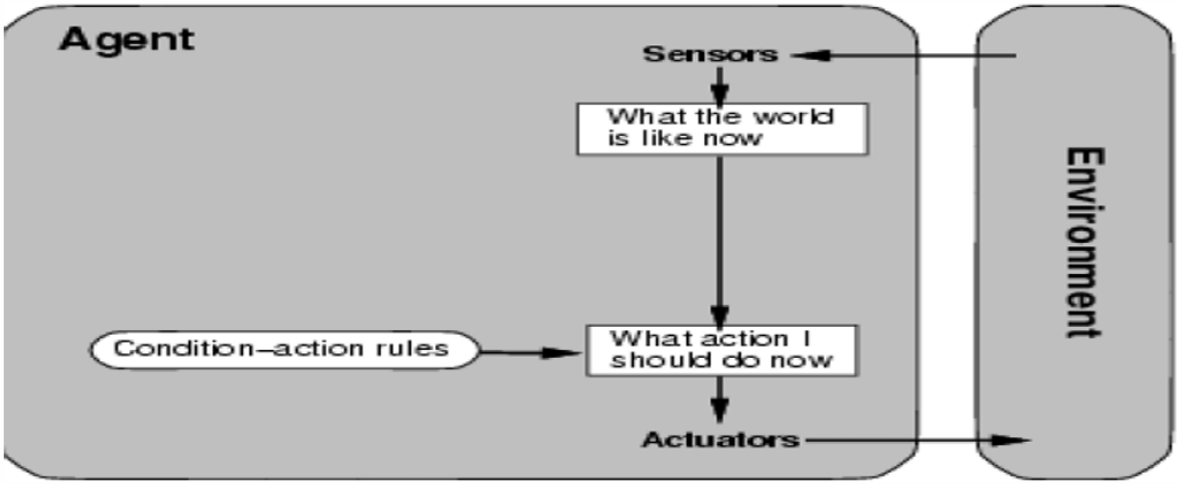
1. Simple Reflex Agent

-> Definition:

Acts only based on the current situation without thinking about the past; it follows simple "if this, then that" rules.

- Example:

A vacuum cleaner that moves if it senses dirt under it.



Advantages:

-Easy to implement

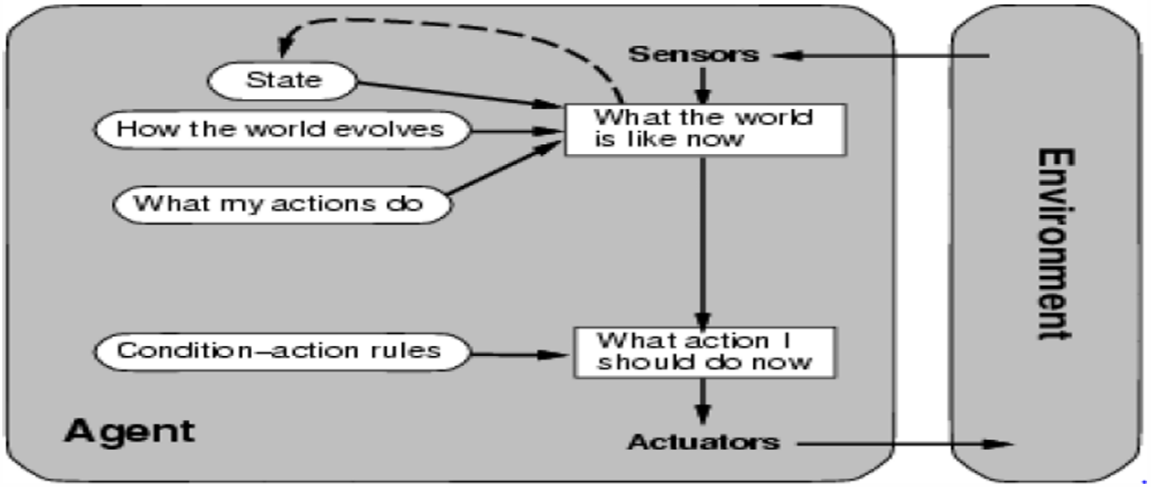
Disadvantage:

-only works on fully observable environment

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2. Model-Based Reflex Agent

Model Based Agent



-> Definition:

Uses some memory (internal model) to remember past events and predict how the world works before deciding what to do.

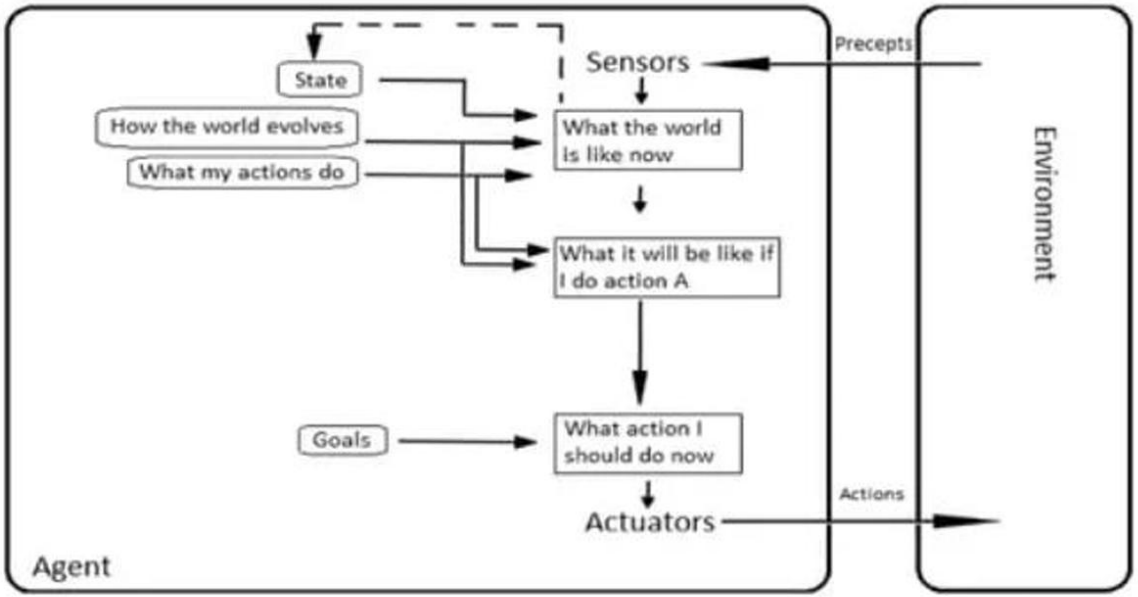
- Example:

A robot that remembers where it already cleaned, even if it doesn't see the dirt now.

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3. Goal-Based Agent

Goal Based Agent



-> Definition:

Thinks about the goal it needs to achieve and chooses actions that move it closer to that goal.

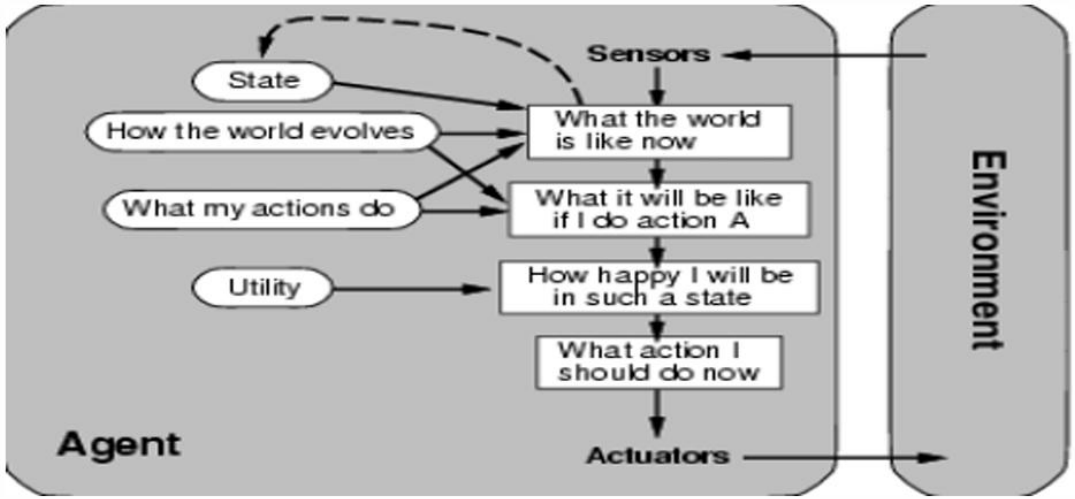
- Example:

A GPS navigation system that finds the best route to your destination.

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4. Utility-Based Agent

Utility Based Agent



-> Definition:

Chooses the best action by calculating how happy or useful the result will be, not just reaching a goal.

- Example:

A self-driving car that not only reaches a destination but also tries to pick the safest, fastest, and most comfortable route.

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5. Learning Agent

-> Definition:

Improves its behavior over time by learning from past experiences and adjusting itself.

-Example:

A spam filter that gets better at catching spam emails as it sees more examples

1.4.5 Environment Types: Deterministic, Stochastic, Static, Dynamic, Observable, Semi-observable

In artificial intelligence, environments are categorized based on several key dimensions that dictate how an agent perceives and interacts with its surroundings.

. Observable vs. Semi-observable

➔ **Fully Observable:** The agent's sensors have complete access to the entire state of the environment at any given time, requiring no internal state to track history.

- **Real-world Example:** Chess is a fully observable environment because both players can see the entire board and all pieces at all times.

➔ **Semi-observable (Partially Observable):** The agent only has access to a portion of the environment due to sensor limitations or hidden variables.

- **Real-world Example:** Autonomous driving is partially observable because the vehicle cannot see around corners or through buildings.

Deterministic vs. Stochastic

➔ **Deterministic:** The next state of the environment is completely determined by the current state and the agent's action, with no randomness involved.

- **Real-world Example:** Solving a Rubik's Cube is deterministic because every twist produces a perfectly predictable rearrangement of stickers.

➔ **Stochastic:** The next state is uncertain and can vary even with the same action, often involving elements of chance or external unpredictability.

- **Real-world Example:** The Stock Market is stochastic because price movements are affected by countless unpredictable global events.

Static vs. Dynamic

➔ **Static:** The environment does not change while the agent is deliberating or "thinking" about its next move.

- **Real-world Example:** Crossword Puzzles represent a static environment; the letters already filled in do not move or change while the player decides on the next word.

➔ **Dynamic:** The environment continues to evolve independently of the agent's actions, often forcing the agent to act under time pressure.

- **Real-world Example:** Playing Soccer is dynamic because opponents and the ball are constantly moving while a player is deciding where to run or pass

Examples:

1. Deterministic Environment

- **Example**: **Chess Game**
    - The outcome of a move is completely predictable — no randomness is involved.

2. Stochastic Environment

- **Example**: **Stock Market Trading**
    - The outcome of an investment decision is uncertain due to market volatility and unpredictable events.

3. Static Environment

- **Example**: **Crossword Puzzle**
    - The puzzle does not change while the agent is solving it.

4. Dynamic Environment

- **Example**: **Autonomous Drone Navigation**
    - The environment changes continuously — people, objects, or weather can affect its path in real time.

5. Fully Observable Environment

- **Example**: **Sudoku Puzzle**

- The agent (solver) has access to all information required — all numbers are visible.

6. Partially Observable (Semi-observable) Environment

- **Example**: **Hide and Seek Game**
  - The seeker cannot see where the hider is — only partial information is available.

Single Agent system and Multi agent system

In AI, a single-agent system involves one autonomous entity handling tasks independently, while a multi-agent system comprises multiple AI agents working together to solve problems. Multi-agent systems excel in dynamic, complex environments due to their adaptability and ability to divide tasks among specialized agents. Single-agent systems are more efficient for focused, well-defined tasks.

➔ **Single-Agent Systems:**

These systems use a single AI entity to process information, make decisions, and perform tasks. They are well-suited for straightforward, well-defined problems where a single agent can handle all aspects.

➔ **Multi-Agent Systems:**

These systems involve multiple AI agents that can collaborate, communicate, and coordinate their actions to achieve a common goal. This approach is particularly effective in complex and dynamic environments where a team of specialized agents can handle different aspects of a problem.

➔ **Advantages of Multi-Agent Systems:**

- **Adaptability:** MAS can adjust to changing environments and requirements more easily than single-agent systems.
- **Specialization:** Different agents can be assigned specific tasks, allowing for optimized performance.
- **Parallel Processing:** Tasks can be divided and processed concurrently by multiple agents, improving efficiency.
- **Robustness:** If one agent fails, others can continue to operate, increasing the system's resilience.

➔ **When to Choose Single vs. Multi-Agent Systems:**

- **Single-Agent:** Use for simple, well-defined tasks where a single, all-encompassing model is suitable.
- **Multi-Agent:** Choose for complex, dynamic environments, tasks that benefit from specialization, and situations requiring collaboration.

Real world example of single agent and multi agent:

Single-agent systems use one AI to complete a task independently, while multi-agent systems use several specialized AIs that collaborate to solve complex problems.

➔ Single Agent Examples (One Solo AI)

- **Smart Thermostat (e.g., Nest):** A single agent that reads room temperature and decides when to turn heat on or off based on settings.
- **Spam Filter**: A solo algorithm that analyzes incoming emails to decide if they belong in the inbox or the spam folder.
- **Chess AI**: A single program that evaluates board positions to choose the best next move against a human opponent.

➔ **Multi-Agent Examples (A Team of AIs)**.

- **Self-Driving Cars** (e.g., Waymo): A system with separate AI agents for detecting pedestrians, navigation, and traffic management working together.
- **AI Software Development Team** (e.g., MetaGPT): Specialized agents for researching, coding, reviewing, and testing code collaborating to create software